



Kickoff Meeting
2026-03-27



Outline

- Motivation
- Project Goal
- Scalable Ethernet-based Network-on-Chip
- Work Packages
- Project Timeline

Motivation

- Scalability limits in FPGA MPSoC designs
- Challenges observed in WireGuard FPGA project
- Traditional NoCs lack interoperability with external networks
- Need for modular, packet-based interconnect

Project Goal

- Develop Ethernet Layer-2 based NoC
- Enable scalable and modular MPSoC architectures
- Unify on-chip and off-chip communication
- Deliver full open-source HW/SW stack

Scalable Ethernet-based Network-on-Chip

- Packet-switched NoC using Ethernet framing
- Switch-based fabric (not shared medium)
- Support for multiple topologies
- Integration with RISC-V and accelerators

Work Packages

- WP1: RTL Gateware Development
- WP2: Integration HAL and Software API Development
- WP3: Verification Infrastructure Development
- WP4: Demo SoCs and Educational Kits Development
- WP5: FOSS Synthesis & PnR Toolchains Integration
- WP6: Repository & Documentation
- WP7: Results Dissemination

WP1 - RTL Gateway Development

This work package delivers the complete openENOC RTL implementation forming the hardware datapath of the Ethernet-based Network-on-Chip. The development progresses from basic switching primitives and datapath elements (RTL1), through packet processing pipeline components (RTL2), to forwarding engines and DMA-capable endpoint connectivity (RTL3). The final stage integrates all components into a configurable Ethernet Layer-2 switching fabric suitable for MPSoC integration (RTL4).

- RTL1 - Core Switching Primitives
- RTL2 - Packet Processing Pipeline
- RTL3 - Forwarding & DMA Engines
- RTL4 - Integrated NoC Fabric

WP2 - Integration HAL and Software API Development

This work package provides the software interface layer required to configure and control the NoC. It begins with the definition of the hardware abstraction layer and register-level interface (SW1), followed by integration with representative RISC-V platforms (SW2). Runtime configuration capabilities are then implemented through a compact baremetal C/C++ API (SW3), while the final stage delivers reference applications demonstrating practical NoC usage scenarios (SW4).

- SW1 - HAL Architecture Specification
- SW2 - RISC-V Platform Integration
- SW3 - Runtime Control API
- SW4 - Reference Applications

WP3 - Verification Infrastructure Development

This work package establishes the verification infrastructure required to validate both the hardware and software components of the platform. It begins with the creation of a simulation environment using Verilator, cocotb and related tools (DV1). Layered testbenches are then developed to verify core datapath modules and packet processing logic (DV2). Hardware/software interaction is validated through co-simulation with instruction set simulation (DV3), and finally continuous regression testing is deployed through automated CI pipelines (DV4).

- DV1 - Simulation Framework
- DV2 - Module-Level Verification
- DV3 - HW/SW Co-Verification
- DV4 - Continuous Verification Infrastructure

WP4 - Demo SoCs and Educational Kits Development

This work package demonstrates the practical applicability of openENOC through deployable FPGA-based systems. The first stage delivers a minimal reference SoC integrating a RISC-V core and openENOC on a low-cost FPGA board (DEM1). The second stage expands the platform with additional endpoints and external Ethernet connectivity, enabling full system demonstrations that bridge on-chip and off-chip networking (DEM2).

- DEM1 - Initial FPGA NoC Demo
- DEM2 - Extended FPGA Demonstration Platform

WP5 - FOSS Synthesis & PnR Toolchains Integration

This work package ensures that openENOC can be built using a fully open-source FPGA toolchain. Initial work integrates the RTL design with synthesis tools such as Yosys and openXC7 (PNR1). The final stage establishes automated place-and-route flows and CI-driven bitstream generation using openXC7/nextpnr, enabling reproducible builds and continuous integration (PNR2).

- PNR1 - Open Synthesis Flow
- PNR2 - Automated Bitstream Build Flow

WP6 - Repository & Documentation

This work package produces the technical documentation and developer resources necessary to make the project accessible and reusable by the wider open hardware community. Initial architecture documentation describing the design concepts and packet model is produced first (DOC1). Detailed design documentation of RTL modules and interfaces follows (DOC2). Tutorials and developer guides are then prepared to demonstrate simulation and FPGA deployment workflows (DOC3), and the final stage documents reproducible build and verification processes (DOC4).

- DOC1 - Initial Architecture Documentation
- DOC2 - Design Documentation
- DOC3 - Tutorials & Developer Guides
- DOC4 - Reproducible Build & Verification Documentation

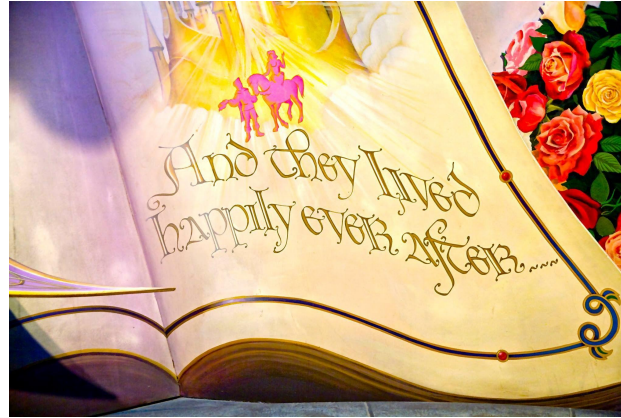
WP7 - Results Dissemination

This work package focuses on communicating the results of the project to the broader open hardware and research communities. The work includes preparation of a technical publication describing the architecture and implementation of openENOC (DIS1) as well as presentation of the project results at relevant open hardware or FPGA-related events (DIS2).

- DIS1 - Technical Publication
- DIS2 - Conference Presentation

Project Timeline

Task / Subtask	Month	1	2	3	4	5	6	7	8	9	10	11	12
		Apr '26	May '26	Jun '26	Jul '26	Aug '26	Sep '26	Oct '26	Nov '26	Dec '26	Jan '27	Feb '27	Mar '27
WP1 - RTL Gateway Development													
1a. RTL1 - Core Switching Primitives		x	x										
1b. RTL2 - Packet Processing Pipeline				x	x								
1c. RTL3 - Forwarding & DMA Engines						x	x						
1d. RTL4 - Integrated NoC Fabric								x	x				
WP2 - Integration HAL and Software API Development													
2a. SW1 - HAL Architecture Specification		x	x										
2b. SW2 - RISC-V Platform Integration				x	x								
2c. SW3 - Runtime Control API						x	x						
2d. SW4 - Reference Applications								x	x				
WP3 - Verification Infrastructure Development													
3a. DV1 - Simulation Framework		x	x										
3b. DV2 - Module-Level Verification				x	x								
3c. DV3 - HW/SW Co-Verification						x	x						
3d. DV4 - Continuous Verification Infrastructure								x	x	x	x		
WP4 - Demo SoCs and Educational Kits Development													
4a. DEM1 - Initial FPGA NoC Demo								x	x				
4b. DEM2 - Extended FPGA Demonstration Platform										x	x	x	x
WP5 - FOSS Synthesis & PnR Toolchains Integration													
5a. PNR1 - Open Synthesis Flow									x	x	x		
5b. PNR2 - Automated Bitstream Build Flow												x	x
WP6 - Repository & Documentation													
6a. DOC1 - Initial Architecture Documentation		x	x										
6b. DOC2 - Design Documentation				x	x	x	x						
6c. DOC3 - Tutorials & Developer Guides								x	x	x	x		
6d. DOC4 - Reproducible Build & Verification Documentation												x	x
WP7 - Results Dissemination													
7a. DIS1 - Technical Publication											x	x	x
7b. DIS2 - Conference Presentation											x	x	x



Thank you!